

Samesun Singh

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PROFILE

I am currently pursuing a Master's degree in Data Science and Engineering at Politecnico di Torino, where I also completed my Bachelor's degree. I have a strong foundation in Python and SQL, with solid analytical and problem-solving skills. I am comfortable working in international environments and communicate effectively in both Italian and English. I am motivated to apply my skills to real-world, data-driven challenges.

SKILLS

Programming languages: Python, SQL, Java

Big Data: Hadoop, Apache Spark

ML Tools & Frameworks: Scikit-learn, Pandas, NumPy, Matplotlib, XGBoost, SHAP, Jupyter, RAG, LangChain, LangGraph

Database: Microsoft SQL Server, Oracle PL/SQL

Other: Git/GitHub, Agile, Docker, Office 365

EXPERIENCE

Thesis Student

Mar '26 — Present

Column news Torino, Italy

- Currently developing a recommendation system to solve the cold start problem for new users and items.
- Researching and implementing machine learning techniques to improve personalized recommendations in sparse-data environments.
- Working with user-item interaction datasets to analyze recommendation patterns and system performance.

Research Student

Sep '25 — Jan '26

Stellantis Torino, Italy

- Designed and implemented a comprehensive ML solution to predict vehicle breakdown risk using over 1 million historical service logs, warranty claims, and sensor signals.
- Built automated data-cleaning and feature-engineering pipelines.
- Trained and compared multiple models (CatBoost, LightGBM, Random Forests).

PROJECTS

ML techniques to predict post-operative complications, LINKS Foundation & Molinette Hospital

Oct '25 — Jan '26

Torino, Italy

- Developed a machine learning system to predict post-operative complications and nosocomial infection risks in ENT oncology patients.

Temporal Localization for NLQ on Egocentric Videos with VideoQA Extension

Feb '25 — Jul '25

Torino, Italy

- Developed and compared span-based models (VSLBase, VSLNet) for temporal localization of natural language queries on Ego4D egocentric videos using Omnivore and EgoVLP features.
- Extended the best model to a VideoQA pipeline with Video-LLaVA and evaluated answers using BLEU and ROUGE metrics.

Regression Model for Age Prediction, Politecnico di Torino

Oct '24 — Jan '25

Torino, Italy

- Developed regression models to predict a speaker's age from extracted linguistic and acoustic features of spoken sentences.

EDUCATION

Master's in Data Science and Engineering, Politecnico di Torino

Oct '24 — Present

Torino, Italy

Bachelor's in Computer Engineering, Politecnico di Torino (GPA: 95/110)

Sep '20 — Sep '24

Torino, Italy

Exchange Semester in Computer Engineering, Mississippi State University (GPA: 4.0/4.0)

Aug '23 — Dec '23

Starkville, United States

PUBLICATION

Microkernel vs. Monolithic OS

Dec '23

Analysed the impact of OS architectures on functional safety, reliability, and fault tolerance in safety-critical systems.

Available at: <https://www.researchgate.net/profile/Samesun-Singh>

LANGUAGES

English (C2), Italian (B2), Spanish (A1), Hindi (C2)